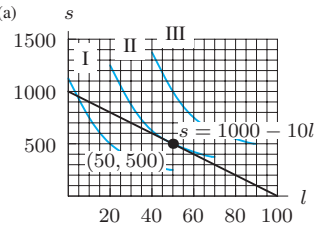
23 (a)



(b) $s = 1000 - 10l$

25 (a) Min; max at endpt of constraint
λ neg
(b) Max; min at endpt of constraint
λ pos

27 $\Delta c/4$; $-\Delta c/4$

31 (a) $C = \$4349$
(b) \$182

33 (a) $W = 225$
$K = 37.5$
(c) $W = 225$
$K = 37.5$
$\lambda = 0.29$

35 (a) No
(b) Yes
(c) $a + b = 1$

37 $x_1 = ((v_1)^{1/2} + (v_2)^{1/2})/(m(v_1)^{1/2})$
$x_2 = ((v_1)^{1/2} + (v_2)^{1/2})/(m(v_2)^{1/2})$

39 (a) $f_1 = \frac{k_1}{k_1+k_2}mg$, $f_2 = \frac{k_2}{k_1+k_2}mg$
(b) Distance the mass stretches the top spring and compresses the lower spring

43 (a) Cost of producing quantity u when prices are p, q
(b) $2\sqrt{pqu}$

45 (a) $-5\lambda^2 + 15\lambda$
(b) 1.5, 11.25
(c) 11.25, 1.5
(d) same

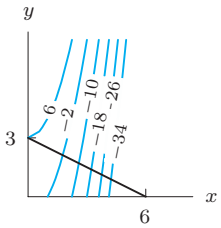
47 (a) $S = \ln(a^a(1-a)^{(1-a)}) + \ln b - a\ln p_1 - (1-a)\ln p_2$
(b) $b = e^c p_1^a p_2^{(1-a)}/(a^a(1-a)^{(1-a)})$

49 Maximum value is 1

51 $f(x, y) = 3x + 4y$

53 $f(x, y) = x^2 + y^2$

55



59 False

61 False

63 False

65 False

67 True

69 True

Chapter 15 Review

1 $(3, -1)$, Saddle point

3 Saddle pt: $(0, -5)$
Local min: $(2, -5)$

5 $(2, 1)$; local min

7 $(2, 3)$: Local and global max

9 Local minimum: $(1, 2)$
Local maximum: $(-1, -2)$
Saddle points: $(1, -2)$ and $(-1, 2)$
No global maximum or minimum

11 Minimum $f(-3/\sqrt{5}, 4/\sqrt{5}) = -5\sqrt{5}$,
Maximum $f(3/\sqrt{5}, -4/\sqrt{5}) = 5\sqrt{5}$

13 Min $= 11.25$; no max

15 Minimum $f(27.907, 23.256) = 1860.484$;
No maximum

17 Maximum $= f(48, 52) = 37{,}600$
No minimum

19 Max $= 21$, no min

21 No global extrema.

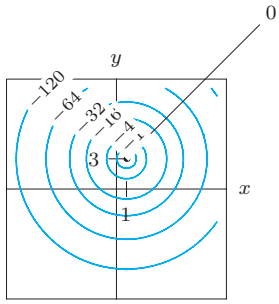
23 Maximum

25 Both

27 0.204

29 $\Delta c/2$; $\Delta c/(2\sqrt{2})$

31

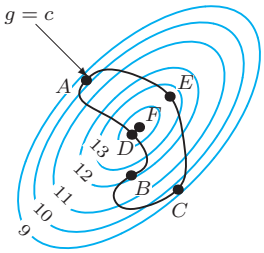


33 (a) (i) Power function
(ii) Linear function
(b) $\ln N = 1.20 + 0.32\ln A$
Agrees with biological rule

35 (a) $2c^2/3$
(b) $4c/3$
(c) $\lambda = m'(c)$

37 $c\left(\frac{aK}{(a+b)P_1}\right)^a\left(\frac{bK}{(a+b)P_2}\right)^b$

39 (a) Points A, B, C, D, E
(b) Point F
(c) Point D



41 $f(x, y, z) = \frac{|Aa+Bb+Cc+D|}{\sqrt{A^2+B^2+C^2}}$

43 $K = 20$
$L = 30$
$C = \$7{,}000$

47 Base: 4 cm by 4 cm
Height: 2 cm

49 $-1/2$

51 (a) $x = 1/4 - a$, $y = 1$, saddle point

53 (a) 21.0208
(b) $g = 20.5, 21.2211$; $g = 20.2, 21.1008$
(c) $g = 20.5, 21.2198$; $g = 20.2, 21.1007$

Section 16.1

1 24; 43.5

3 Over: Approx 137
Under: Approx 60

5 about 2300

7 Positive

9 Zero

11 Zero

13 Positive

15 25.2°C

17 Need f nonnegative everywhere

19 $f(x, y) = 5 - x - y$; R is square with vertices $(\pm 1, \pm 1)$

21 False

23 False

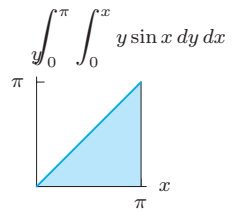
25 True

27 True

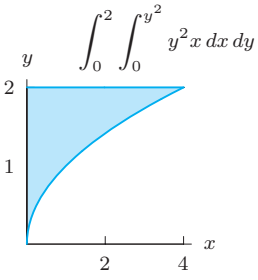
29 False

Section 16.2

1



3



5 150

7 54

9 $e - 2$

11 $3 - \sin 3$

13 $(e^4 - 1)(e^2 - 1)e$

15 -2.678

17 $\int_1^4 \int_1^2 f\,dy\,dx$ or $\int_1^2 \int_1^4 f\,dx\,dy$

19 $\int_{-1}^3 \int_{-2}^{(1-3x)/4} f\,dy\,dx$
or $\int_{-2}^1 \int_{-1}^{(1-4y)/3} f\,dx\,dy$